Victorine Younang, Ph.D.

GenAI / LLM Engineer — Ph.D., Computer Science

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SUMMARY

Machine Learning Engineer and CS Ph.D. specializing in large language models — fine-tuning, adaptation, and rigorous evaluation. Fine-tuned and benchmarked six open LLMs (CodeLlama, StarCoder2, DeepSeek-Coder, Qwen2.5, and a DeepSeek-V2 MoE) on a single 16 GB GPU using LoRA/QLoRA, PEFT, and 4-bit quantization, with custom evaluation metrics beyond BLEU/CodeBLEU. Shipped production ML — a graph recommender at Rec Room (+33% session time, −10% latency). Builds retrieval-augmented (RAG) applications with the OpenAI API and ChromaDB, and owns the full model lifecycle: data → fine-tuning/adaptation → evaluation → containerized deployment. Published in IEEE TDSC and IEEE TSC. Security-grounded: publications in top security venues (IEEE TDSC, IEEE TSC), an NSF project on generative AI for secure systems, and verification-first ("trustworthy AI") systems — a strong fit for AI-safety / model-evaluation / red-teaming and AI-security roles.

TECHNICAL SKILLS

- **AI Safety & Security:** LLM evaluation & safety/robustness metrics (custom), AI red-teaming / dangerous-capability-style evaluation, trustworthy & verified AI (execution-verified code generation), adversarial ML, secure/trusted code generation, backdoor detection (CipherGuard), formal verification (Isabelle/HOL), AST-based static analysis, Bayesian attack graphs & threat modeling (CVSS); OWASP Top 10 for LLMs (familiar); security research in IEEE TDSC/TSC with an NDSS submission; NSF "Generative AI for secure systems and software" contributor.
- **LLMs & GenAI:** LLM fine-tuning (LoRA, QLoRA, PEFT), instruction tuning, model distillation, Mixture-of-Experts (MoE), curriculum learning, RAG, OpenAI API, vector search / retrieval with ChromaDB (and Qdrant), prompt engineering, LLM evaluation & custom metrics, BERT/transformer fine-tuning, agentic systems (LangGraph).
- **Frameworks & Training:** PyTorch, Hugging Face (Transformers, PEFT), PyTorch Geometric, bitsandbytes; 4-/8-bit quantization (NF4), gradient checkpointing & accumulation, Distributed Data Parallel (DDP), GPU memory optimization; vLLM serving.
- **ML & Deep Learning:** Graph Neural Networks (GNN/GAT), representation learning, NLP, classification, anomaly detection, class-imbalance & small-data methods.
- **Deployment & MLOps:** Docker, Kubernetes, Azure Cloud AI, Git/GitHub, model serving & latency optimization, Weights & Biases / MLflow.
- **Data & Languages:** Python, SQL/NoSQL, Pandas, NumPy, scikit-learn, ETL pipelines, large-scale scraping (Selenium/BeautifulSoup); R.

EXPERIENCE

Oakland University — Graduate Research Assistant (ML / LLMs)

Michigan, USA · Sep 2020 – Apr 2026

- Built a curriculum fine-tuning framework adapting six open LLMs for a specialized code-translation task; achieved 96.9% syntactic validity with strong zero-shot generalization to unseen inputs.
- Engineered the training stack for a single 16 GB GPU (LoRA/QLoRA, 4-bit NF4 quantization, gradient checkpointing/accumulation, label-masked PEFT), making multi-model experiments feasible on constrained hardware.
- Designed custom LLM evaluation metrics (syntactic, semantic, and value correctness) beyond BLEU/CodeBLEU.
- Authored/co-authored 6 papers (3 peer-reviewed & published, 3 under submission); peer reviewer (5 manuscripts, 2024); primary contributor on the NSF project "Using Generative AI to Design and Develop Secure Systems and Software."
- Instructor of record (30+ students); TA across Deep Learning, Databases, and Networks.

Rec Room Inc. — Applied ML Research Intern

Seattle, USA · May 2023 – Aug 2023

- Built and productionized a graph-based recommender (random-walk + Node2Vec-style embeddings) over 1,000+ environments; +33% average session time on first deployment.
- Refactored the real-time inference pipeline in Python for a 10% latency reduction; added cold-start logic from structural graph properties.

EDF (Électricité de France) — Junior Big Data Consultant

Paris, France · Sep 2019 – Sep 2020

- Technical-functional lead for a consent-management platform (millions of profiles); engineered GDPR data-erasure on Hadoop (100% compliance); cut downstream errors 30% via standardized validation pipelines.

Orange Telecom — Data Science Intern
Brussels, Belgium · May 2019 – Aug 2019

- Built multivariate time-series models for network capacity planning; Python ingestion pipelines (Kudu/Elasticsearch); internal R/Hadoop package cut query-setup overhead 15%.

SELECTED PROJECTS

- **RepoSage — Agentic code-&-docs RAG with execution-verified answers** (LangGraph, vLLM, Qdrant, DPO): dual-source RAG over a real library's code + docs; for code tasks it generates → executes in a sandbox → self-corrects against the library's own tests. Self-hosted Qwen2.5-Coder via vLLM (4-bit); DPO alignment from execution-feedback preferences; RAGAS/Pass@1 evaluation harness. (See projects/reposage.md.)
- **RAG Study Assistant** (OpenAI API, ChromaDB — hackathon): upload papers → chunk → embed → store vectors → answer from retrieved context; a full ingest → retrieve → generate loop.
- **TIA — Curriculum Fine-Tuning across 6 LLMs** (LoRA/QLoRA, MoE, PyTorch, Hugging Face): imperative→formal code translation on a single GPU; 3 custom correctness metrics; best syntax validity 0.969 with 0.76–0.92 zero-shot generalization. Under submission, AAAI 2027.
- **CipherGuard — reference-free backdoor detection for cipher implementations** (first-author; under submission, NDSS 2027): static analysis that flags test-vector-compliant backdoors in lightweight block ciphers via a "data-oblivious invariant" — analyzed 44 formally-verified ciphers with **zero false positives**, catching a branchless key-leak backdoor that constant-time analyzers and known-answer tests miss; AST + GNN (GAT) representation with an Isabelle/HOL-verified corpus of 414 labeled implementations across Feistel/ARX/SPN families. (Model/code integrity & ML-supply-chain security.)
- **CryptoGraph — Graph ML for Structural Classification** (GAT+MLP, PyTorch Geometric; under submission, IEEE TIFS): hybrid GNN over AST graphs + global features; F1 = 0.94 with zero-shot generalization, beating GCN, Transformer-Conv, RF, XGBoost; attention-based interpretability.
- **Security Risk Assessment with Bayesian Attack Graphs & Complex Probabilities** (first-author; under submission): novel use of complex probabilities to quantify attack likelihood/impact and model destructive interference + backtracking cycles in BAGs; CVSS v3.1-based; validated on a Connected & Autonomous Vehicles network; open-source.
- **Provable-Security Framework for IoT** (Ph.D. dissertation): multi-tiered design, optimization, and formal verification of resource-constrained IoT ecosystems — from perceived trust to provable security.
- **NLP at Scale — BERT Sentiment/Stance Classifier**: scraped 19,000+ posts, fine-tuned BERT to 96% accuracy, with topic modeling over time. Springer LNCS (2022).

CERTIFICATIONS (IN PROGRESS)

- **AWS Certified AI Practitioner (AIF-C01)** — in progress, expected Nov 2026; AWS AI/ML stack incl. Amazon Bedrock and SageMaker.
- **NVIDIA-Certified Associate: Generative AI & LLMs (NCA-GENL)** — in progress, expected Dec 2026; transformer architectures, prompt engineering, fine-tuning, and NVIDIA NeMo / Triton / TensorRT.
- **OWASP Top 10 for LLM Applications** — studying; the de-facto LLM-security framework (prompt injection, insecure output handling, model DoS, supply-chain), demonstrated via RepoSage.

EDUCATION

- **Ph.D., Computer Science** — Oakland University, Michigan (Conferred Apr 2026)
- Dissertation: *From Perceived Trust to Provable Security: A Multi-Tiered Framework for the Design, Optimization, and Verification of Resource-Constrained IoT Ecosystems.*
- **M.Sc. Big Data & B.Sc. Computer Engineering** — ESIGELEC, France (2015–2019)

SELECTED PUBLICATIONS

- **Peer-reviewed:** IEEE TDSC (2025); IEEE TSC (2025) — *Extending IoT Devices Network Lifetime* (VAR forecasting + QoS recommender; 95% accuracy, +20% network lifetime); Springer ICIOT/LNCS (2022).
- **Under submission:** *CipherGuard* — reference-free cipher-backdoor detection (NDSS 2027); *CryptoGraph* — learning security properties of block ciphers with graph ML (IEEE TIFS, resubmission); *Security Risk Assessment with Bayesian Attack Graphs & Complex Probabilities*; *TIA* — curriculum fine-tuning across 6 LLMs (AAAI 2027).